

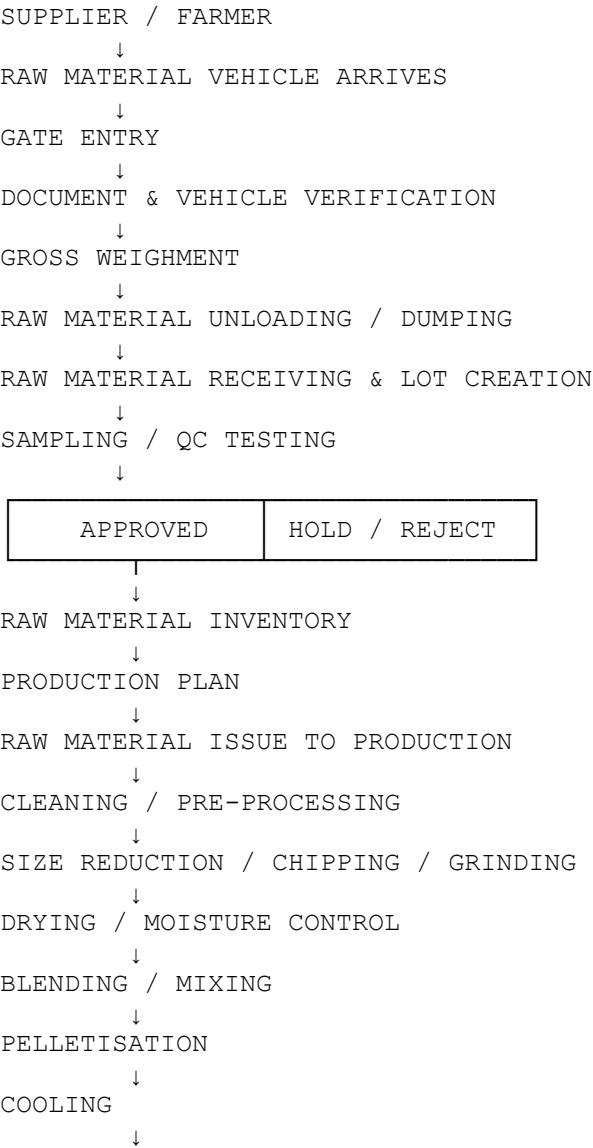
Every Raw Material → Batch → Production Run → Finished Goods Batch → Dispatch
should remain digitally traceable.

MAHAURJA – PLANT OPERATIONAL FLOW

Biomass Pellet Manufacturing Management System

Bharat Industrial & Renewables LLP

1. MASTER OPERATIONAL FLOW





2. DETAILED SOFTWARE WORKFLOW

STAGE 1 — Supplier / Farmer

Before material arrives, the system should have a **Supplier/Farmer Master**.

Supplier information

- Supplier ID
- Supplier name
- Mobile number
- Village
- Tehsil
- District
- Material supplied
- Bank details, where required
- GST/PAN, where applicable
- Supplier type:
 - Farmer
 - Trader
 - Aggregator

- Company
- Historical purchase quantity
- Historical purchase rate
- Payment history
- Active/Inactive status

Example

Supplier ID: SUP-000145

Name: Ramesh Patel

Village: XYZ

Material: Groundnut Shell

Mobile: XXXXX XXXXX

3. RAW MATERIAL VEHICLE ARRIVAL

When the vehicle reaches the factory, the **Gate/Security module** starts the transaction.

Gate Entry

System generates:

RM-GATE-261002-001

Enter:

- Date
- Time
- Vehicle number
- Driver name
- Driver mobile
- Supplier
- Material
- Purpose
- Purchase Order, if applicable
- Expected quantity
- Security remarks

Vehicle status

ARRIVED

↓

GATE ENTRY CREATED

↓

WAITING FOR WEIGHMENT

The system should show the vehicle on the **Gate Dashboard**.

4. GROSS WEIGHMENT

Vehicle goes to the weighbridge.

System records:

- Vehicle number
- Gross weight
- Date
- Time
- Weighbridge slip number
- Operator
- Photo/document, if required

Example:

Gross Weight = 28,450 kg

Vehicle status:

Gross Weighed

5. RAW MATERIAL UNLOADING

Vehicle is directed to the appropriate unloading/storage location.

Example:

- RM Yard A – Groundnut Shell
- RM Yard B – Cashew Shell
- RM Yard C – Wood
- RM Yard D – Agro Residue

The system should record:

Unloading Location / Storage Location

This is important because the management needs to know **where the physical material is actually stored**.

6. SAMPLING & QUALITY CONTROL

Before material becomes fully usable inventory, the QC/Lab team takes a sample.

QC parameters

Depending on material:

- Moisture %
- Ash %
- GCV
- Foreign Matter %
- Bulk Density
- Material Grade
- Other applicable parameters

System generates:

QC-261002-001

QC status

PENDING

↓

TESTING

↓

APPROVED / HOLD / REJECTED

Important software rule

Rejected material should NOT automatically become available production inventory.

Similarly:

HOLD material should remain blocked until authorised release.

7. TARE WEIGHTMENT

After unloading, the empty vehicle returns to the weighbridge.

System records:

Tare Weight = 10,200 kg

The software automatically calculates:

Net Quantity = Gross Weight – Tare Weight

Example:

28,450 – 10,200 = 18,250 kg

Therefore:

Accepted Quantity

18.250 MT

The system should not allow the operator to manually calculate this every time.

8. RAW MATERIAL RECEIPT

Now the system creates the actual **Raw Material Receipt / Purchase Transaction**.

Example:

RM-261002-001

Linked records:

Supplier
↓
Vehicle
↓
Gate Entry
↓
Gross Weighment
↓
QC Report
↓
Tare Weighment
↓
Net Quantity
↓
Purchase Transaction
↓
Inventory Lot

This linkage is extremely important.

9. RAW MATERIAL LOT CREATION

Every accepted raw material should receive a **Lot/Batch ID**.

Example:

RMLOT-GS-261002-001

Where:

- GS = Groundnut Shell
- 261002 = 2 Oct 2026
- 001 = Lot number

The lot should contain:

- Material
- Supplier
- Vehicle
- Date received
- Quantity
- Purchase rate
- QC report
- QC parameters
- Storage location
- Status

10. RAW MATERIAL INVENTORY

The inventory system should now increase stock.

Example:

Groundnut Shell

Opening Stock: 125 MT

Received: +18.25 MT

Available: **143.25 MT**

But the system should also know:

Which 18.25 MT came from which supplier, vehicle and QC lot?

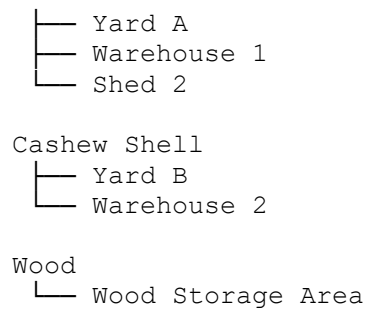
11. RAW MATERIAL STORAGE

The system should support **location-wise inventory**.

Example:

RAW MATERIAL INVENTORY

Groundnut Shell



This becomes important when the plant grows.

12. PRODUCTION PLANNING

Management/Production Supervisor creates a **Production Plan**.

Example:

Production Plan: PRD-261002-001

Date: 02/10/2026

Shift: Morning

Target: 60 MT

Product: MAHAURJA Biomass Pellet

Diameter: 8 mm

The production plan may specify a particular **blend/formula**.

Example:

Material	Planned Quantity
Groundnut Shell	30 MT
Cashew Shell	15 MT
Wood	15 MT
Total	60 MT

The exact formula should be configurable rather than hard-coded.

13. RAW MATERIAL ISSUE TO PRODUCTION

Production requests material from the raw-material inventory.

System creates:

RM ISSUE – ISS-261002-001

Example:

- Groundnut Shell: 30 MT
- Cashew Shell: 15 MT
- Wood: 15 MT

Inventory automatically decreases.

RAW MATERIAL STOCK
↓
MATERIAL ISSUE
↓
PRODUCTION

The system must record **which raw-material lots were consumed**.

This enables complete traceability.

14. PRE-PROCESSING

The production process begins.

Depending on the plant machinery/process:

Stage A — Cleaning

Remove:

- stones
- foreign matter
- unwanted material

System can record:

- Input quantity
- Output quantity
- Rejected quantity

- Loss
-

15. SIZE REDUCTION / GRINDING

Raw material is processed to the required particle size.

System can record:

- Machine
 - Start time
 - End time
 - Input quantity
 - Output quantity
 - Operator
 - Production shift
 - Machine downtime
-

16. DRYING / MOISTURE CONTROL

Material is dried/conditioned to the required moisture level.

Record:

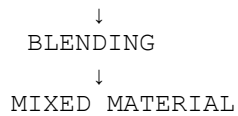
- Input quantity
 - Output quantity
 - Moisture before
 - Moisture after
 - Dryer used
 - Energy consumption, if captured
 - Loss
-

17. BLENDING / MIXING

Different materials may be blended according to the production formula.

Example:

```
Groundnut Shell
+
Cashew Shell
+
Wood
```



The system should capture:

- Formula ID
- Raw-material lots
- Actual quantities
- Target quantities
- Operator
- Production batch

18. PELLETISATION

The blended material enters the pellet mill.

Record:

- Production batch
- Machine
- Shift
- Start time
- End time
- Input quantity
- Output quantity
- Pellet diameter
- Operator
- Downtime
- Remarks

Example:

Production Batch: PB-261002-001

Product:

MAHAURJA Biomass Pellet – 8 mm

19. COOLING

Fresh pellets pass through the cooling process.

The software can track:

- Batch
 - Input
 - Output
 - Start/end time
 - Machine
 - Operator
 - Loss
-

20. SCREENING

Pellets are screened to separate:

- Finished pellets
- Fines
- Oversize/undesirable material

System records:

Good Production

Fines

Rejected/Recycle Quantity

21. FINISHED GOODS BATCH

After production, the system creates a **Finished Goods Batch**.

Example:

FG-BATCH-261002-001

Information:

- Product
 - Batch number
 - Production date
 - Production quantity
 - Pellet diameter
 - Formula
 - Raw-material lots consumed
 - Production shift
 - Machine
 - Operator
-

22. FINAL QUALITY CONTROL

Finished pellets go through final laboratory testing.

Possible parameters

- GCV
- Moisture
- Ash
- Bulk density
- Pellet diameter
- Fines %
- Other customer-specific parameters

System generates:

FG-QC-261002-001

Status:

PENDING

↓

TESTING

↓

APPROVED

OR

HOLD

OR

REJECTED

Again:

Only approved finished goods should become dispatchable inventory.

23. PACKAGING

The system should support multiple packaging modes.

Option A — Bagged

Example:

- 25 kg bag
- 40 kg bag
- 50 kg bag
- Customer-specific packaging

Option B — Bulk / Loose

Pellets dispatched directly in bulk vehicle.

System records:

- Batch
 - Packaging type
 - Number of bags
 - Bag weight
 - Total quantity
 - Packaging material used
 - Packaging date
-

24. FINISHED GOODS STORAGE

Approved finished pellets move into the finished goods warehouse/storage area.

Example:

FG WAREHOUSE

8 mm Premium Pellet

↓

Batch FG-BATCH-261002-001

↓

100 MT

Inventory:

Finished Goods Available = 100 MT

The software must distinguish:

Produced → QC Pending → QC Approved → Dispatchable

25. SALES ORDER

Customer places an order.

Example:

SO-261002-015

Customer:

ABC Industries Pvt. Ltd.

Order:

- Product: MAHAURJA Biomass Pellet
- Diameter: 8 mm
- Quantity: 50 MT
- Rate: ₹____ / MT
- Delivery location
- Required date
- Payment terms
- Customer specification

Status:

NEW

↓

CONFIRMED

↓

PARTIALLY DISPATCHED

↓

FULLY DISPATCHED

26. DISPATCH PLANNING

Once the order is approved, the dispatch team creates a dispatch plan.

System checks:

Customer requirement

50 MT

Available approved FG

100 MT

Available to dispatch

50 MT

Then:

Dispatch Approved

27. DISPATCH VEHICLE ARRIVAL

Vehicle enters factory.

System records:

- Vehicle number
- Driver name
- Driver mobile
- Transporter
- Customer
- Sales Order
- Dispatch number
- Arrival time

Example:

DIS-261002-001

28. LOADING

The system links the loading transaction to:

```
Customer
  ↓
Sales Order
  ↓
Dispatch
  ↓
Finished Goods Batch
  ↓
Vehicle
```

This is critical.

The system should know:

Which exact production batch was sent to which customer in which vehicle?

29. DISPATCH WEIGHMENT

Vehicle is weighed.

Example:

Gross:

27,850 kg

Tare:

12,850 kg

Net:

15,000 kg = 15 MT

System automatically calculates net dispatch quantity.

30. INVOICE & DOCUMENTATION

System generates/records:

- Sales Invoice
 - Delivery Challan, if applicable
 - E-way bill, where applicable
 - Weighbridge slip
 - LR/transport document
 - QC/COA, where applicable
 - Customer PO
 - Other required documents
-

31. VEHICLE EXIT

Security verifies:

- Vehicle
- Quantity
- Invoice
- Dispatch documents
- Customer
- Authorisation

Then:

Vehicle Exit Completed

Finished goods inventory automatically reduces.

Example:

Before dispatch:

100 MT

Dispatch:

15 MT

Balance:

85 MT

32. CUSTOMER DELIVERY

Dispatch status can progress:

DISPATCHED
↓
IN TRANSIT
↓
DELIVERED
↓
POD RECEIVED

Where required, upload:

- Delivery confirmation
 - POD
 - Customer acknowledgement
 - Weighment slip
 - Quality feedback
-

33. PAYMENT

The sales transaction then connects to accounts.

Example:

SALES ORDER
↓
DISPATCH
↓
INVOICE
↓
RECEIVABLE
↓
PAYMENT

↓
RECEIPT

The system should show:

Invoice Amount: ₹ _____

Received: ₹ _____

Outstanding: ₹ _____

Due Date: _____

34. THE MOST IMPORTANT PART — TRACEABILITY

This is something I strongly recommend making a **core architectural requirement** for the developers.

A single customer delivery should be traceable backwards.

Example:

Customer received 15 MT

↓

Dispatch **DIS-261002-001**

↓

Finished Goods Batch **FG-BATCH-261002-001**

↓

Production Batch **PB-261002-001**

↓

Production Formula

↓

Raw Material Lots

↓

Supplier/Farmer

↓

Vehicle

↓

QC Report

↓

Weighbridge Records

Therefore:

Customer → Dispatch → FG Batch → Production Batch → Raw Material Lot → Supplier

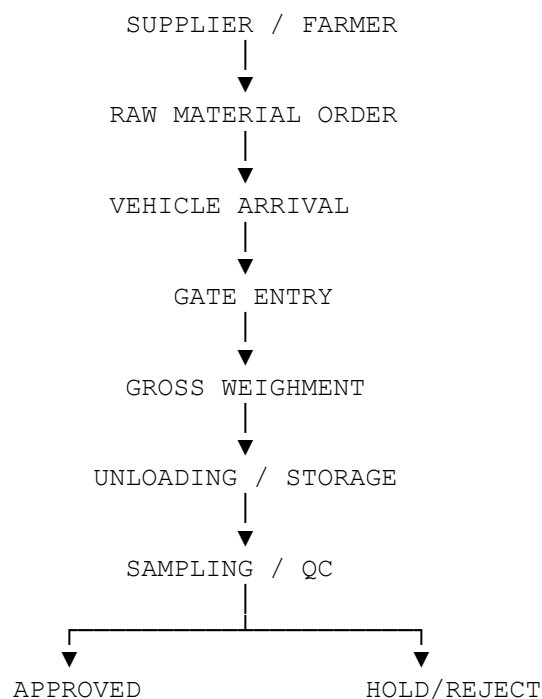
And the reverse should also work:

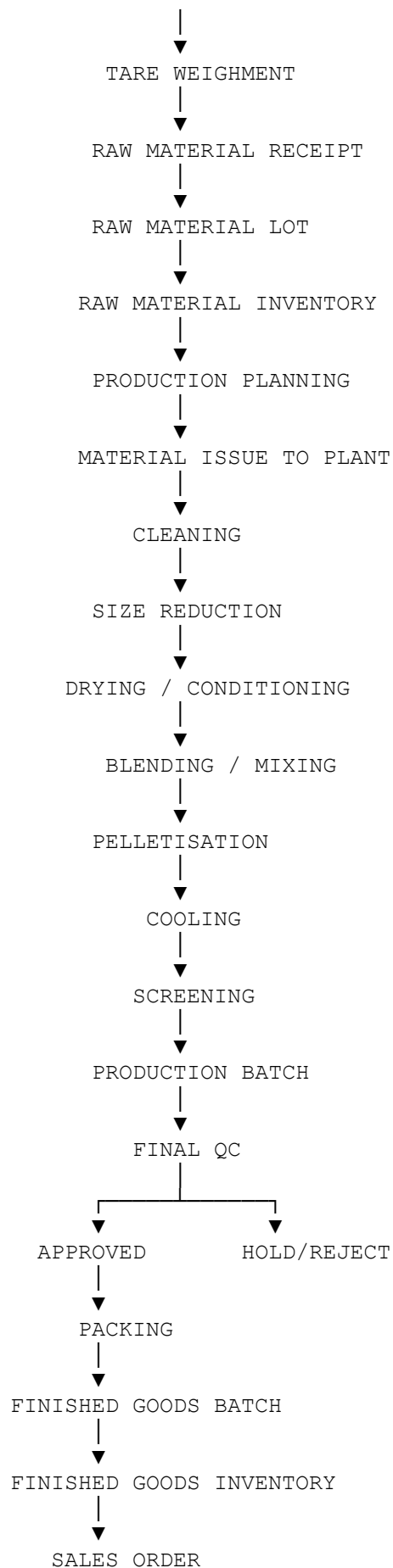
Supplier → Raw Material Lot → Production Batch → Finished Goods Batch → Customer

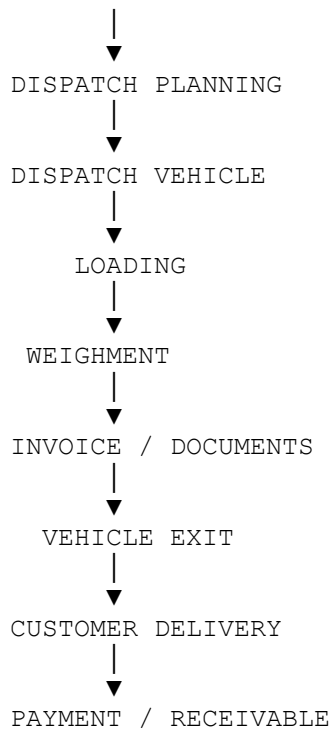
This will become extremely valuable as MAHAURJA grows.

35. COMPLETE SYSTEM FLOW

I would give your developers this as the **master flow**:







36. SOFTWARE STATUS SYSTEM

I would also ask the developers to create **status-based workflows**, rather than simply entering data into forms.

Raw Material

Expected → Arrived → Gross Weighed → Unloaded → QC Pending → Approved/Hold/Rejected → Tare Weighed → Received → Stored → Issued → Consumed

Production

Planned → Material Issued → Processing → Pelletisation → Cooling → Screening → Produced → QC Pending → Approved/Hold/Rejected → Stored

Finished Goods

Produced → QC Pending → Approved → Available → Reserved → Loading → Dispatched → Delivered

Sales Order

Enquiry → Quotation → Order Received → Confirmed → Production/Stock Reserved → Partially Dispatched → Fully Dispatched → Closed

Payment

Invoice Generated → Outstanding → Part Payment → Fully Paid → Closed

37. DASHBOARD SHOULD SHOW THIS FLOW LIVE

The management dashboard should be able to show something like:

Area	Live Status
Vehicles Inside Plant	4
Raw Material Awaiting QC	2
Raw Material Available	385 MT
Production Today	72 MT
Production Target	100 MT
FG Awaiting QC	18 MT
Finished Goods Stock	245 MT
Orders Pending	180 MT
Dispatches Today	3
Dispatch Quantity	65 MT
Outstanding Receivables	₹XX
Raw Material Average Cost	₹XX/MT
Production Cost	₹XX/MT

This turns the software into a **plant operating system**, rather than merely an accounting program.

38. One rule I would make non-negotiable

For every major transaction, the developers should ask:

“What happened physically, and what corresponding digital transaction should happen?”

For example:

Truck physically arrives

→ Gate Entry

Truck physically gets weighed

→ Weighment Entry

Material physically gets unloaded

→ Receiving Entry

Material physically gets tested

→ QC Entry

Material physically enters inventory

→ Stock Entry

Material physically enters production

→ Material Issue

Pellets physically come out

→ Production Batch

Pellets physically pass QC

→ FG Approval

Pellets physically enter warehouse

→ FG Stock

Pellets physically leave factory

→ Dispatch

Customer physically receives pellets

→ Delivery/POD

Customer physically pays

→ Receipt

That principle will prevent the software from becoming a collection of disconnected screens and will give MAHAURJA **complete operational, inventory, quality, cost and traceability control.**